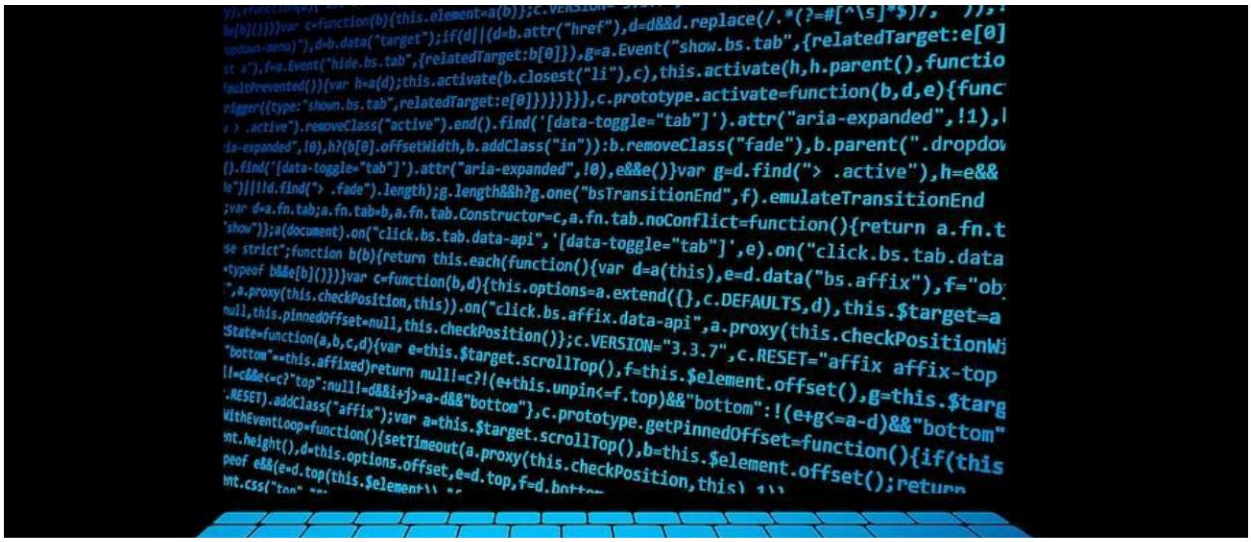
Al-Aqsa University

Faculty of Information Technology

Information Retrieval

جـامعـة الأقـصـى



Text Search Engine using TF-IDF

Haitham Hani Dalloul

24/01/2026

Instructor:
Mohammed Jamil Altahrawi

2025-2026
First Semester

Introduction

Information Retrieval is the process of fetching relevant information for the user's query from a large dataset. TF-IDF (Term Frequency–Inverse Document Frequency) estimates the significance of terms within a document included in a set of documents. By representing documents and queries as numerical vectors using TF-IDF and cosine similarity, the system provides a ranked list of documents that best match the user's query.

Dataset Description

The dataset used in the project from scikit-learn named 20 news groups. it contains a good content for the TF-IDF processing, more than the dataset contains a large number of documents , but the first 30 documents are selected for the project, and changed the data into CSV format easier manipulation and processing.

Text Preprocessing

1. Tokenization: split the text into words, because this stage is important since the nltk library works with split words, and it is easier to work with words than with text.

2. Lowercasing: convert all text to lowercase because the TF-IDF model considers uppercase and lowercase words as different.

3. Stop words: remove them because they can reduce the accuracy of the TF-IDF results.

4. Punctuation: remove it because they do not add meaning to the text.

5. Digits: delete numbers because they are not important for text-based retrieval, and most people search with words rather than numbers.

6. Remove any special signs because they do not have meaning in search.

7. Stemming: reduce words to their root form to avoid giving every word a separate weight, such as 'used' and 'using,' which are the same in meaning. Using their root helps in better weighting.

8. Join words together: restore the words to form the document because the search will be conducted on the document, not on a single word.

9. White space removal: Remove white spaces to search faster and to ensure a clean and consistent document representation.

TF-IDF Model

TF-IDF (Term Frequency–Inverse Document Frequency) is a common method used for displaying text documents numerically.

TF represents the weight of the word in the same document, while the IDF represents how a term among the documents set is unique or rare.

Each document is represented as a numerical vector where each dimension corresponds to a term weight.

These weights allow the system to compare the relevance of the term to the user query using Cosine similarity.

In Python, `TfidfVectorizer` from the `sklearn` library can be used to efficiently convert a collection of documents into TF-IDF vectors for further analysis.

Search and Ranking

After preprocessing, the user query applies the same steps applied to documents, including tokenization, stop word removal, punctuation removal, digit removal, sign removal, stemming, and joining words.
The queries are ready and have now converted into a TF-IDF vector to calculate the cosine similarity between the query and each document vector.

Cosine similarity measures the angle between two vectors, indicating how well a document matches the query. The documents are ranked in descending order of similarity, and the top five most relevant are returned.

Analysis & Discussion

• **Why did the top-ranked document achieve the highest similarity score?**

The top-ranked document achieves the highest similarity because it contains terms that are in the query user and have high TF-IDF weights. These terms are important and unique within the document set, which increases the cosine similarity between the query and the document. That's why this document is ranked higher than others.

• **Which terms contributed most to the similarity score?**

The terms that appear in the user query and the document and have high TF-IDF weights are the most affected in the cosine similarity; in this project, 'footbal' 'championship' and 'result' are the most important. These terms are unique and important across the document set, which increases the cosine similarity calculation, therefore helping the document to get a higher ranking.

• **What is the effect of removing stop words?**

Removing stop words has a good effect on the search result, as they are common words that have barely any semantic meaning, and keeping them makes noise in the TF-IDF result. By removing them, the system focuses on terms that are more important and unique in both query and documents, which leads to improved accuracy of cosine similarity, and relevant documentsare ranked higher.

Additional Note: When I keep stop words in the documents and user query affects TF-IDF and cosine similarity results because these common words add noise, which may reduce the accuracy of similarity scores.

Conclusion

The project successfully demonstrates how TF-IDF and cosine similarity can be applied to build a basic yet effective text search engine, fulfilling the core objectives of information retrieval.